

BluDrop: Offline Bluetooth Mesh Communication System

1 Project Overview

BluDrop is an Android-based offline communication application designed to enable device-to-device messaging without requiring an internet connection. The system uses Bluetooth-based peer-to-peer communication to exchange messages between nearby devices.

The project focuses on enabling communication in environments where internet connectivity is unavailable, unreliable, or intentionally avoided.

2 Working of the System

The communication process in BluDrop follows these steps:

1. A user creates and sends a message from the application.
2. A unique UUID is assigned to each message. This UUID acts as a unique identifier for tracking the message across devices.
3. The message is transmitted to nearby devices using Bluetooth communication.
4. When a device receives a message, it checks whether the message UUID has already been processed.
5. If the UUID has already been seen, the message is treated as a duplicate and is not processed or rebroadcast.
6. If the UUID is new, the message is accepted, stored locally, and may be forwarded to other nearby devices.
7. This enables multi-hop communication, where intermediate devices can relay messages to devices outside the direct Bluetooth range of the sender.

3 Duplicate Suppression

In mesh-style communication, the same message may reach a device through multiple paths. Without duplicate suppression, devices may repeatedly process and rebroadcast the same message.

BluDrop addresses this problem by assigning a unique UUID to every message. Each device maintains a record of previously processed message UUIDs.

- If a received UUID has already been seen, the message is discarded.
- If the UUID is new, the message is processed and may be forwarded to nearby devices.

This mechanism reduces redundant message transmissions and prevents repeated re-broadcasting.

4 Multi-Hop Communication

A key feature of BluDrop is multi-hop communication. Two devices do not necessarily need to be directly connected to exchange messages.

For example:

Device A $\rightarrow$ Device B $\rightarrow$ Device C $\rightarrow$ Device D

If Device A cannot directly communicate with Device D, intermediate devices can act as relays and forward the message. This allows communication to extend beyond the direct Bluetooth range.

5 Use Cases

BluDrop can be useful in situations where traditional internet-based communication is unavailable or unreliable.

- **Disaster and Emergency Communication:** Communication when mobile networks or internet infrastructure is unavailable.
- **Remote Locations:** Messaging in rural, mountainous, or remote areas with limited network connectivity.
- **Crowded Events:** Communication during concerts, festivals, or large gatherings where mobile networks may become congested.
- **Campus and Local Communities:** Offline communication between users within a limited geographical area.
- **Temporary Network Outages:** Maintaining local communication when internet connectivity is temporarily unavailable.

6 Current System Limitations

The current implementation is a prototype and has certain limitations. Bluetooth communication range is limited, and message delivery depends on the availability of nearby intermediate devices for multi-hop forwarding.

As the number of participating devices increases, message propagation, duplicate tracking, network congestion, and scalability become important challenges.

Future improvements may include optimized routing mechanisms, controlled message forwarding, improved device discovery, and more scalable message tracking.